Software Engineering Project Jan-2026

Milestone 1 Deliverables for Cargo-Core

Submitted By

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Team Code : **Team-003**

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Small Business Operations Platform for Rural Logistics Management

Project Details

Software Name: Cargo-Core

Department Context: Local-Logistic-Department

System Type: Small Business Logistics & Resource Management Platform

Date of Submission : 22-02-2026

Team Details

Team Name: QuadCore-Devs
Team Code: Team-003

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Identification of Users

1 Primary Users:

These are the core operational users of the system.
The platform’s workflows, dashboards, and real-time coordination features will be primarily designed around their daily logistics execution responsibilities.
Any disruption in their usage directly impacts delivery completion and warehouse

operations. ● **Warehouse Manager (e.g., Raghavendra)**

Handles physical inventory inside the godown and ensures readiness for dispatch.
Responsibilities include:

- Organizing storage of mixed goods (grains, fertilizers, furniture, construction materials)
- Locating items quickly for dispatch
- Monitoring packing materials such as boxes and tape
- Issuing and tracking returnable equipment (crates, belts, blankets)
- Coordinating with dispatcher during loading preparation

The system must support structured inventory tracking and equipment accountability to reduce time wastage and losses.

● **Dispatcher (e.g., Mohd Yusuf)**

Acts as the operational coordinator responsible for matching orders to resources. Responsibilities include:

- Assigning appropriate vehicles for each order
- Allocating correct number of labourers
- Scheduling deliveries during peak and non-peak seasons
- Managing delays and breakdown situations
- Coordinating between warehouse, driver, and owner

The system must assist in resource allocation, reduce miscalculations, and improve scheduling visibility.

● **Driver (Field Executor)**
Responsible for on-ground transportation and delivery execution. Responsibilities include:

- Receiving job details and destination information
- Transporting goods safely across rural routes
- Updating delivery status

- Handling loading/unloading coordination at site
- Ensuring return of issued equipment
- Communicating delays caused by route or network issues

The system must provide clear job details, route guidance, and delivery status recording to reduce disputes and fuel inefficiency.

2. Secondary Users:

These users interact with the system for monitoring, booking, and financial management. They are essential stakeholders but are not involved in physical execution of logistics tasks.

● Owner (Logistics Manager – e.g., Kallappa Biradar)

The strategic overseer of the entire logistics operation. Responsibilities include:

- Monitoring bookings and order volume
- Tracking payments and pending dues
- Supervising payroll calculations
- Reviewing fuel expenses and vehicle usage
- Managing peak season demand pressure
- Ensuring overall profitability and operational stability

The system must provide dashboard visibility, financial tracking, and operational summaries to reduce manual stress.

● Customer (Farmer / Vendor / Individual Client – e.g., Shifa)

External service requester who depends on the logistics system for timely goods transportation. Responsibilities include:

- Placing booking requests
- Providing accurate details of goods
- Making full or partial payments
- Tracking delivery status
- Receiving confirmation of completed service

The system must provide structured booking, confirmation receipts, and transparency to reduce miscommunication and delays.

3. Tertiary Users:

These users are part of the logistics workflow but do not directly interact with the technology platform. Their activities are recorded and managed by primary users.

● Labourers / Helpers

The physical workforce responsible for loading and unloading goods at warehouse and delivery sites. Their characteristics include:

- Receiving work assignments verbally
- Working variable hours depending on order size
- Assisting drivers in heavy lifting tasks
- Handling equipment such as belts and crates

In the system design, their attendance, work allocation, and equipment usage are managed through dispatcher or driver entries rather than direct system login.

Users Interviewed During Research:

We conducted interviews with the following stakeholders:

- Kallappa Biradar – Owner (Logistics Manager)
- Raghavendra – Warehouse Manager
- Mohd Yusuf – Dispatcher
- Driver – Field Executor
- Labourer – Helper
- Shifa – Customer

USER RESEARCH & PAIN POINT ANALYSIS

(Based on interviews with Owner, Customer, Warehouse Manager, Dispatcher and Driver)

1. Research Objective

To understand operational challenges, coordination gaps, and systemic inefficiencies in a rural logistics business operating in Shirashyad through stakeholder interviews.

2. Key Observational Insights

Observation 1: Fully Manual System

All operations (booking, billing, payroll, warehouse management, dispatching) are handled manually through phone calls and paper records.

Observation 2: Owner as Central Node

The owner acts as the sole communication bridge between customers, drivers, dispatcher, and warehouse. This creates dependency and operational overload.

Observation 3: No Data Visibility

There is no system to track:

- Orders
- Payments
- Vehicles
- Equipment
- Working hours

This leads to confusion, disputes, and inefficiency.

3. Stakeholder Pain Points

3.1. Owner

- ₹50,000 pending dues from customers
- Manual invoice generation
- Manual payroll calculation
- Suspected fuel bill manipulation
- Peak season demand clashes
- No live tracking of vehicles

Core Issue: Financial leakage and operational stress due to lack of monitoring tools.

3.2. Customer

- Verbal booking only (no confirmation receipt)
- Difficulty explaining goods quantity over phone
- No delivery tracking
- Cost-driven underestimation of vehicle size
- Delays during harvest season affecting income

Core Issue: Lack of transparency and booking guidance.

3.3. Warehouse Manager

- No systematic storage layout
- Difficulty locating goods
- Packing material shortages discovered last minute
- No tracking of returnable equipment

Core Issue: Unorganized inventory and asset management.

3.4. Dispatcher

- Manual vehicle and labour assignment
- Miscalculation of required resources
- Large delays when jobs exceed expected time
- No structured contingency planning

Core Issue: No decision-support system for allocation and scheduling.

3.5. Driver

- Incomplete job details
- No route optimization
- Network dead zones causing communication issues
- No proof-of-delivery mechanism
- No work-hour tracking

Core Issue: Inefficient field coordination and navigation.

3.6. Labourer

- No fixed schedule
- Frequent underestimation of labour requirement
- No attendance or work-hour record
- Weekly pay based on rough estimates
- No equipment accountability

Core Issue: Informal workforce management and payroll opacity.

4. Cross-Stakeholder Patterns

Recurring issues across multiple roles:

- Manual communication dependency
- Resource miscalculation
- Equipment loss
- No tracking system
- No work-hour documentation
- Payment ambiguity
- Peak-season operational overload

5. Root Causes Identified

- Absence of structured booking system
- No centralized information platform
- No tracking of vehicles or equipment
- No inventory monitoring
- No payroll automation
- No route planning system

6. Overall Conclusion

The business demand is stable and seasonal.
However, inefficiencies arise due to lack of structure, visibility, and accountability mechanisms.

The logistics service currently operates on a trust-based, phone-driven model.

System-level digitization is required to improve coordination, reduce losses, and support scalability.

User Stories

(The following user stories follow the S.M.A.R.T guidelines (Specific, Measurable, Achievable, Relevant, Time-bound) Format: As a [User], I want [Action], So that [Benefit].)

Epic 1: Order Booking & Confirmation

Story 1.1: Structured Booking Request

- **As a** Customer,
- **I want** to submit a structured booking request with details of goods, quantity, and location
- **So that** the service provider clearly understands my requirements without confusion

Story 1.2: Booking Confirmation Receipt

- **As a** Customer,
- **I want** to receive a digital confirmation after booking a vehicle
- **So that** I have written proof of my order details and service cost

Story 1.3: Cost & Resource Suggestion

- **As a** Customer,
- I **want** the system to suggest the appropriate vehicle size and number of labourers - **So that** I avoid underestimating resources and prevent delivery delays

Epic 2: Payment & Billing Management

Story 2.1: Digital Invoice Generation

- **As a** Owner,
- **I want** the system to automatically generate invoices for every completed order - **So that** I do not have to create manual receipts

Story 2.2: Payment Tracking

- **As a** Owner,

- **I want** to track full, partial, and pending payments
- **So that** I can monitor outstanding dues and reduce revenue loss

Story 2.3: Flexible Payment Recording

- **As a** Customer,
- **I want** to record partial payments and remaining balance clearly -
So that both parties have transparency in transactions

Epic 3: Fleet & Route Management

Story 3.1: Optimized Route Planning

- **As a** Driver,
- **I want** the system to provide optimized navigation routes
- **So that** I reduce fuel consumption and avoid unnecessary backtracking

Story 3.2: Live Vehicle Tracking

- **As a** Owner,
- **I want** to see real-time location of vehicles
- **So that** I can monitor deliveries without constantly calling drivers

Story 3.3: Delivery Status Updates

- **As a** Customer,
- **I want** to receive delivery status updates
- **So that** I do not need to repeatedly call for information

Epic 4: Warehouse & Inventory Management

Story 4.1: Structured Storage Layout

- **As a** Warehouse Manager,
- **I want** goods to be stored with mapped sections and labeled locations
- **So that** items can be located quickly during dispatch

Story 4.2: Inventory Level Alerts

- **As a** Warehouse Manager,
- **I want** automatic alerts when packing materials or stock runs low - **So that** I can restock before dispatch delays occur

Story 4.3: Equipment Issue & Return Tracking

- **As a** Warehouse Manager,
- **I want** to record issued and returned equipment for each order
- **So that** losses of crates, belts, and blankets can be minimized

Epic 5: Resource Allocation & Scheduling

Story 5.1: Automated Resource Assignment

- **As a** Dispatcher,
- **I want** the system to recommend suitable vehicles and labour count for each order - **So that** miscalculations and delays are reduced

Story 5.2: Job Schedule Visibility

- **As a** Dispatcher,
- **I want** to generate and share daily schedules for labourers
- **So that** they can prepare for work without last-minute confusion

Story 5.3: Load Time Estimation

- **As a** Dispatcher,
- **I want** estimated loading and unloading time predictions
- **So that** delivery schedules can be managed realistically

Epic 6: Workforce & Payroll Management

Story 6.1: Work Hour Recording

- **As a** Driver,
- **I want** my start and end work time to be recorded digitally
- **So that** my payment calculation is fair and transparent

Story 6.2: Labour Attendance Tracking

- **As a** Owner,
- **I want** to track labour attendance and working hours
- **So that** payroll calculation becomes accurate and stress-free

Story 6.3: Weekly Payroll Summary

- **As a** Owner,
- **I want** the system to generate a weekly payroll summary for each labourer - **So that** I can provide them with transparent payment details and avoid wage disputes

Epic 7: Communication & Coordination

Story 7.1: Central Communication Platform

- **As a** Owner,
- **I want** all communication between customers, drivers, and dispatcher to be recorded in one system
- **So that** misunderstandings and information gaps are reduced

Story 7.2: Offline Status Sync

- **As a** Driver,
- **I want** to update delivery status even in low network areas
- **So that** information is synced once connectivity is restored

Story 7.3: Priority Handling During Peak Season

- **As a** Owner,
- **I wan**t to prioritize orders during peak demand
- **So that** urgent deliveries are handled efficiently

Epic 8: Equipment & Asset Protection

Story 8.1: Equipment Accountability

- **As a** Dispatcher,
- **I want** to digitally confirm equipment received and returned for each job - **So that** the team is not unfairly blamed for missing items

Story 8.2: Fuel Usage Monitoring

- **As a** Owner,
- **I want** to compare vehicle mileage with fuel expenses -
So that fraudulent fuel claims can be identified

Link of Interviews and Proofs of Interaction :- Link